Filippos N. Tzortzoglou

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Summary

Ph.D. in Civil and Environmental Engineering, Cornell University, focusing on motion planning, optimal control, and multi-agent coordination of connected and automated vehicles in mixed traffic. Combines control theory with optimization and machine learning: MPC, control barrier functions, Lyapunov methods, mixed-integer and distributed optimization, and data-driven modeling of human drivers. MathWorks internship (2024).

Education

Cornell University | **Ithaca, NY, USA** August 2023 – December 2026 (exp.)
Ph.D., Transportation Systems Engineering (CAV motion planning and coordination)

Cornell University | **Ithaca, NY, USA** August 2023 – November 2025
M.S., Transportation Systems Engineering

Technical University of Crete | **Chania, Greece** September 2017 – July 2022
Diploma (Integrated Master's) in Production Engineering and Management

Honors and Awards

- **IEEE Outstanding Student Paper Award 2024 on Smart Cities**, awarded at the 64th IEEE Conference on Decision and Control (CDC 2025), Rio de Janeiro, for "A Feasibility Analysis at Signal-Free Intersections" (IEEE LCSS).
- **Award for Excellence in Teaching 2026**, Cornell University Graduate and Professional Student Assembly.
- **Inclusive Graduate Educator of the Year 2026**, Cornell University Office of Inclusive Excellence.
- **Finalist, Best Paper Award**, 17th IFAC Symposium on Control in Transportation Systems (CTS 2024).
- **Student Spotlight**, Cornell Graduate School (February 2026).

Experience

Graduate Researcher, Information and Decision Science Lab August 2023 – Present
Cornell University, Ithaca, NY, USA

- Designed optimal control and coordination algorithms for CAVs at intersections in mixed traffic (MPC, CBFs, mixed-integer and distributed optimization) on two NSF-funded projects.
- Lead the lab's research collaboration with the MathWorks Controls Group through the DCRG program, coordinating regular technical meetings with the Controls Group's engineering leadership.

Engineering Intern, MathWorks Controls Group | **Natick, MA, USA** June 2024 – August 2024

- Developed CAV control algorithms for scenarios with pedestrians in RoadRunner Scenario, co-simulated with MATLAB/Simulink.
- Implemented real-time, sensor-based pedestrian detection and collision avoidance in Simulink.

Research Assistant, Dynamic Systems and Simulation Lab | **TU Crete** July 2021 – July 2022

- Involved in ERC Advanced Grant *TrafficFluid*: decentralized control of autonomous vehicles on lane-free roads.

Teaching Assistant, Cornell University August 2025 – December 2025

- *Uncertainty for Engineers* (students evaluation 4.67/5).

Selected Publications

- **F.N. Tzortzoglou**, L.E. Beaver, A.A. Malikopoulos, "Safe and Efficient Coexistence of Autonomous Vehicles with Human-Driven Traffic at Signalized Intersections", *64th IEEE Conference on Decision and Control (CDC)*, 2025.
- **F.N. Tzortzoglou**, C. Walters, A.A. Malikopoulos, "Data-Driven Modeling and Prediction of Human Drivers in Mixed Traffic Using Virtual Reality", *European Control Conference (ECC)*, 2026.
- **F.N. Tzortzoglou**, L.E. Beaver, A.A. Malikopoulos, "A Feasibility Analysis at Signal-Free Intersections", *IEEE Control Systems Letters (LCSS)*, 2024.
- V. Le, B. Chalaki, **F.N. Tzortzoglou**, A.A. Malikopoulos, "Stochastic Time-Optimal Trajectory Planning for Connected and Automated Vehicles in Mixed-Traffic Merging Scenarios", *IEEE Transactions on Control Systems Technology (TCST)*.
- H. Bang, A. Dave, **F.N. Tzortzoglou**, A.A. Malikopoulos, "On Mobility Equity and the Promise of Emerging Transportation Systems", *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*.

Skills

Methods: MPC, control barrier functions, Lyapunov theory, Pontryagin's minimum principle, MIQP, distributed optimization (ADMM), multi-agent reinforcement learning
Programming: Python, MATLAB/Simulink, C **Tools:** RoadRunner, SUMO, VISSIM, Unity (VR), Gurobi

Leadership and Outreach

- Mentor numerous undergraduate and graduate students at Cornell on research projects in autonomous driving.
- Share research on autonomous vehicles and control with a professional network of 18k+ followers on LinkedIn.